

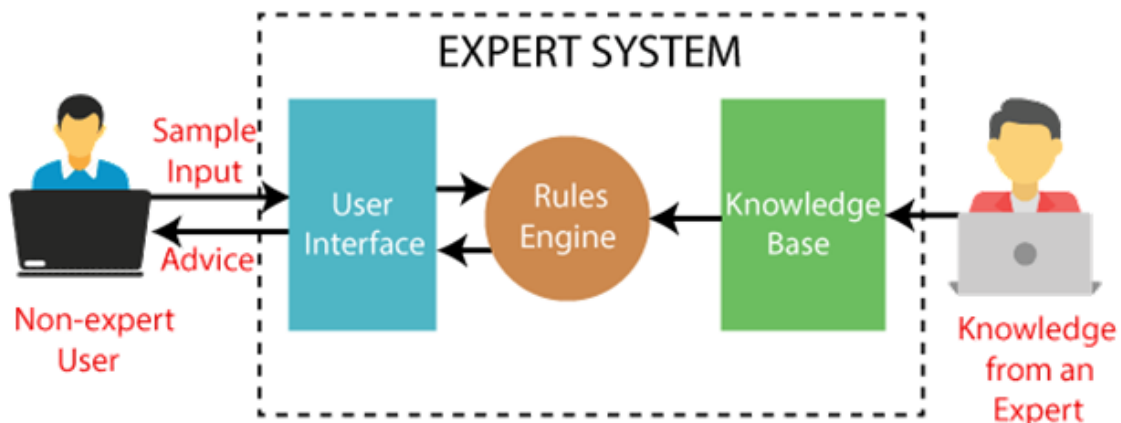
Unit-5: Expert System

Expert System: An expert system is a computer program that is designed to solve complex problems and to provide decision-making ability like a human expert. It performs this by extracting knowledge from its knowledge base using the reasoning and inference rules according to the user queries.

The expert system is a part of AI, and the first ES was developed in the year 1970, which was the first successful approach of artificial intelligence. It solves the most complex issue as an expert by extracting the knowledge stored in its knowledge base. The system helps in decision making for complex problems using both facts and heuristics like a human expert. It is called so because it contains the expert knowledge of a specific domain and can solve any complex problem of that particular domain. These systems are designed for a specific domain, such as medicine, science, etc.

The performance of an expert system is based on the expert's knowledge stored in its knowledge base. The more knowledge stored in the KB, the more that system improves its performance. One of the common examples of an ES is a suggestion of spelling errors while typing in the Google search box.

Below is the block diagram that represents the working of an expert system:



Examples of the Expert System:

DENDRAL: It was an artificial intelligence project that was made as a chemical analysis expert system. It was used in organic chemistry to detect unknown organic molecules with the help of their mass spectra and knowledge base of chemistry.

MYCIN: It was one of the earliest backward chaining expert systems that was designed to find the bacteria causing infections like bacteraemia and meningitis. It was also used for the recommendation of antibiotics and the diagnosis of blood clotting diseases.

PXDES: It is an expert system that is used to determine the type and level of lung cancer. To determine the disease, it takes a picture from the upper body, which looks like the shadow. This shadow identifies the type and degree of harm.

CaDeT: The CaDet expert system is a diagnostic support system that can detect cancer at early stages.

Architecture of an Expert System

An expert system is a computer program that simulates the reasoning and decision-making of a human expert. The architecture of an expert system consists of three main components:

- 1. User Interface:** which is the component that interacts with the user and presents the results of the inference engine.
- 2. Inference Engine:** which is the component that applies logical rules and reasoning to the knowledge base to derive conclusions and recommendations.
- 3. Knowledge Base:** which is the repository of domain-specific knowledge that the expert system relies on.
 - The knowledgebase is a type of storage that stores knowledge acquired from the different experts of the particular domain. It is considered as big storage

of knowledge. The more the knowledge base, the more precise will be the Expert System.

- It is similar to a database that contains information and rules of a particular domain or subject.
- One can also view the knowledge base as collections of objects and their attributes. Such as a Lion is an object and its attributes are it is a mammal, it is not a domestic animal, etc.

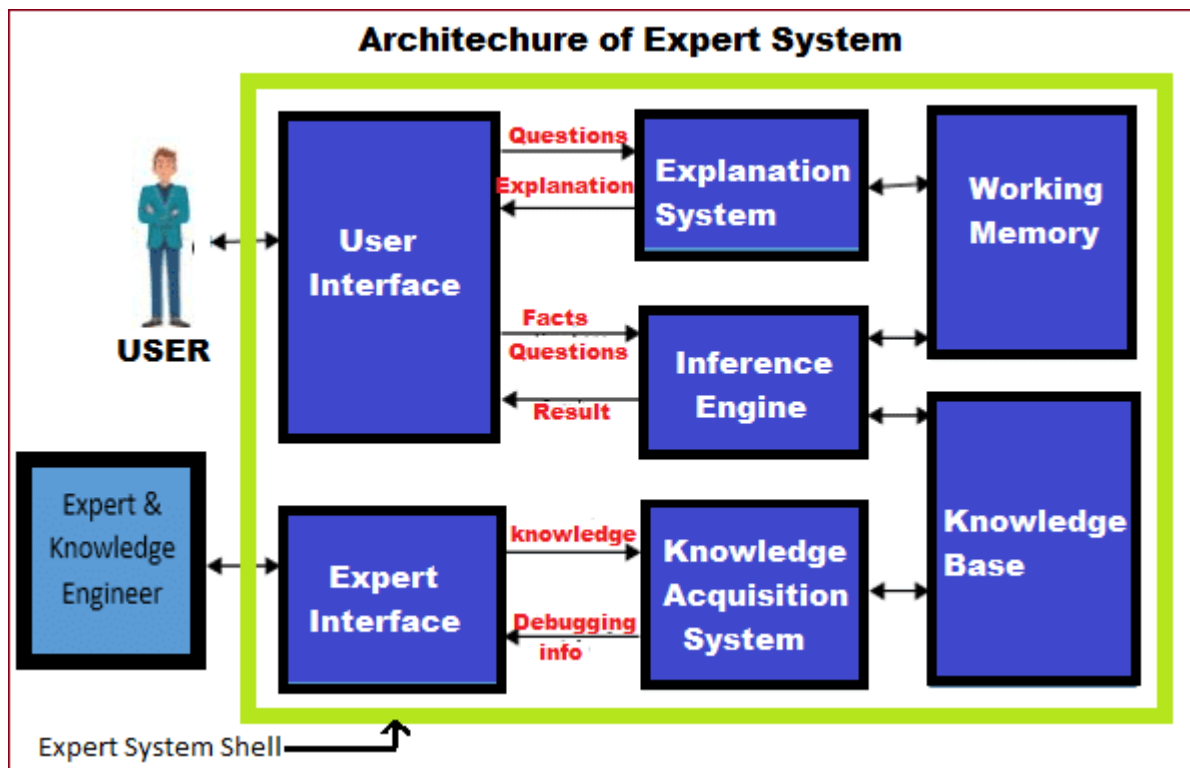


Fig: Architecture of Expert System

Knowledge Representation: It is used to formalize the knowledge stored in the knowledge base using the If-else rules.

Categories of Knowledge:

Knowledge is an awareness of facts, a familiarity with individuals and situations, or a practical skill. Knowledge of facts, also called propositional knowledge, is often

characterized as true belief that is distinct from opinion or guesswork by virtue of justification. Categories of knowledge include:

Explicit knowledge: knowledge that can be easily documented and shared, such as facts, rules, and principles.

Implicit knowledge: knowledge that is learned through practice and experience, such as skills, know-how, and intuition.

Tacit knowledge: knowledge that is difficult to articulate and transfer, such as beliefs, values, and insights.

Declarative knowledge: knowledge that describes what something is, such as definitions, concepts, and classifications.

Procedural knowledge: knowledge that describes how to do something, such as methods, algorithms, and techniques.

Knowledge Acquisitions:

It is the process of extracting, organizing, and structuring the domain knowledge, specifying the rules to acquire the knowledge from various experts, and store that knowledge into the knowledge base.

Knowledge Elicitation:

Knowledge elicitation consists of a set of techniques and methods that attempt to elicit the knowledge of a domain expert, typically through some form of direct interaction with the expert. Knowledge elicitation is a sub-process of knowledge acquisition (which deals with the acquisition or capture of knowledge from any source), and knowledge acquisition is, in turn, a sub-process of knowledge engineering (which is a discipline that has evolved to support the whole process of specifying, developing and deploying knowledge-based systems).

Knowledge Elicitation Techniques:

- **Interviews:** Structured, unstructured, or semi-structured conversations with experts.
- **Think-aloud protocols:** Experts verbalize their thought processes while performing tasks.
- **Observation:** Watching experts perform tasks to understand their implicit knowledge.
- **Questionnaires and surveys:** Collecting information through written responses.
- **Workshops and focus groups:** Collaborative sessions to gather insights from multiple experts.

Development of Expert System

Here, we will explain the working of an expert system by taking an example of MYCIN ES. Below are some steps to build an MYCIN:

- Firstly, ES should be fed with expert knowledge. In the case of MYCIN, human experts specialized in the medical field of bacterial infection, provide information about the causes, symptoms, and other knowledge in that domain.
- The KB of the MYCIN is updated successfully. In order to test it, the doctor provides a new problem to it. The problem is to identify the presence of the bacteria by inputting the details of a patient, including the symptoms, current condition, and medical history.
- The ES will need a questionnaire to be filled by the patient to know the general information about the patient, such as gender, age, etc.

- Now the system has collected all the information, so it will find the solution for the problem by applying if-then rules using the inference engine and using the facts stored within the KB.
- In the end, it will provide a response to the patient by using the user interface.

Participants in the development of Expert System

There are three primary participants in the building of Expert System:

- **Expert:** The success of an ES much depends on the knowledge provided by human experts. These experts are those persons who are specialized in that specific domain.
- **Knowledge Engineer:** Knowledge engineer is the person who gathers the knowledge from the domain experts and then codifies that knowledge to the system according to the formalism.
- **End-User:** This is a particular person or a group of people who may not be experts, and working on the expert system needs the solution or advice for his queries, which are complex.

Why Expert System?

Before using any technology, we must have an idea about why to use that technology and hence the same for the ES. Although we have human experts in every field, then what is the need to develop a computer-based systems. So below are the points that are describing the need of the ES:

No memory Limitations: It can store as much data as required and can memorize it at the time of its application. But for human experts, there are some limitations to memorize all things at every time.

High Efficiency: If the knowledge base is updated with the correct knowledge, then it provides a highly efficient output, which may not be possible for a human.

Expertise in a domain: There are lots of human experts in each domain, and they all have different skills, different experiences, and different skills, so it is not easy to get a final output for the query. But if we put the knowledge gained from human experts into the expert system, then it provides an efficient output by mixing all the facts and knowledge

Not affected by emotions: These systems are not affected by human emotions such as fatigue, anger, depression, anxiety, etc. Hence the performance remains constant.

High security: These systems provide high security to resolve any query.

Considers all the facts: To respond to any query, it checks and considers all the available facts and provides the result accordingly. But it is possible that a human expert may not consider some facts due to any reason.

Regular updates improve the performance: If there is an issue in the result provided by the expert systems, we can improve the performance of the system by updating the knowledge base.

Capabilities of the Expert System

Below are some capabilities of an Expert System:

Advising: It is capable of advising the human being for the query of any domain from the particular ES.

Provide decision-making capabilities: It provides the capability of decision making in any domain, such as for making any financial decision, decisions in medical science, etc.

Demonstrate a device: It is capable of demonstrating any new products such as its features, specifications, how to use that product, etc.

Problem-solving: It has problem-solving capabilities.

Explaining a problem: It is also capable of providing a detailed description of an input problem.

Interpreting the input: It is capable of interpreting the input given by the user.

Predicting results: It can be used for the prediction of a result.

Diagnosis: An ES designed for the medical field is capable of diagnosing a disease without using multiple components as it already contains various inbuilt medical tools.

Advantages of Expert System

- These systems are highly reproducible.
- They can be used for risky places where the human presence is not safe.
- Error possibilities are less if the KB contains correct knowledge.
- The performance of these systems remains steady as it is not affected by emotions, tension, or fatigue.
- They provide a very high speed to respond to a particular query.

Limitations of Expert System

- The response of the expert system may get wrong if the knowledge base contains the wrong information.
- Like a human being, it cannot produce a creative output for different scenarios.
- Its maintenance and development costs are very high.
- Knowledge acquisition for designing is much difficult.
- For each domain, we require a specific ES, which is one of the big limitations.
- It cannot learn from itself and hence requires manual updates.

Applications of Expert System

- In designing and manufacturing domain

It can be broadly used for designing and manufacturing physical devices such as camera lenses and automobiles.

- In the knowledge domain

These systems are primarily used for publishing the relevant knowledge to the users. The two popular ES used for this domain is an advisor and a tax advisor.

- In the finance domain

In the finance industries, it is used to detect any type of possible fraud, suspicious activity, and advise bankers that if they should provide loans for business or not.

- In the diagnosis and troubleshooting of devices

In medical diagnosis, the ES system is used, and it was the first area where these systems were used.

- Planning and Scheduling

The expert systems can also be used for planning and scheduling some particular tasks for achieving the goal of that task.

Case study on Expert Systems

Case study on expert systems, focusing on the application of an expert system in medical diagnosis:

Case Study: MYCIN - Expert System for Medical Diagnosis

Background

MYCIN was an early expert system developed in the 1970s at Stanford University. Its primary purpose was to assist physicians in the diagnosis and treatment of bacterial infections, specifically bacteremia and meningitis. This expert system

represented a significant advancement in the application of artificial intelligence in medicine.

Problem

The diagnosis and treatment of bacterial infections require a deep understanding of microbiology, patient symptoms, and appropriate antibiotic therapies. Physicians often face challenges due to the complexity of infections, variability in patient responses, and the extensive knowledge required to make accurate decisions. In the 1970s, access to specialized knowledge was limited, especially in smaller or less-equipped hospitals.

Objectives

1. To develop a system that could emulate the decision-making abilities of human infectious disease experts.
2. To improve the accuracy and speed of diagnosis and treatment recommendations for bacterial infections.
3. To make specialized knowledge accessible to a broader range of medical practitioners.

System Development

MYCIN was developed using rule-based reasoning, a common approach for expert systems at the time. It utilized a backward chaining inference mechanism to simulate the diagnostic process of human experts. The system comprised several key components:

- **Knowledge Base:** Contained a comprehensive set of rules derived from infectious disease experts. These rules included facts about symptoms, patient history, lab test results, and appropriate antibiotic treatments.
- **Inference Engine:** Used backward chaining to reason from the patient's symptoms and test results to a potential diagnosis and treatment plan.

- **User Interface:** Allowed physicians to input patient data and interact with the system to receive diagnostic recommendations.

Implementation

The implementation of MYCIN involved the following steps:

1. **Knowledge Acquisition:** Collaborating with infectious disease experts to gather the necessary rules and knowledge for the system.
2. **Rule Encoding:** Encoding the acquired knowledge into the system's rule-based framework.
3. **System Testing:** Validating the system's recommendations against actual case studies and expert evaluations.
4. **Refinement:** Continuously updating and refining the rules based on feedback from real-world use.

Results

MYCIN was found to provide recommendations that were often as accurate as those made by human experts. In some cases, the system's suggestions were even more consistent due to its extensive and systematic knowledge base. However, MYCIN was never widely implemented in clinical practice due to several limitations:

- **Complexity of Use:** The system required detailed input from users, which could be time-consuming.
- **Technology Limitations:** The computational resources required were significant for the time.
- **Acceptance:** There was resistance from some medical professionals who were hesitant to rely on a computer-based system for critical decisions.

Impact and Legacy

Despite its limitations, MYCIN had a lasting impact on the field of artificial intelligence and expert systems. It demonstrated the potential of AI to support

complex decision-making processes and paved the way for future developments in medical informatics. Key contributions include:

1. **Rule-Based Systems:** MYCIN validated the effectiveness of rule-based systems in encapsulating expert knowledge.
2. **Knowledge Engineering:** Highlighted the importance of knowledge acquisition and refinement in the development of expert systems.
3. **AI in Medicine:** Inspired subsequent research and development of more advanced medical expert systems and decision support tools.

Conclusion:

MYCIN is a landmark example of how expert systems can be applied to solve complex problems in specific domains. Although it faced challenges in widespread adoption, the insights and methodologies it introduced continue to influence the development of AI-based diagnostic tools and expert systems across various fields. The case study of MYCIN underscores the importance of integrating expert knowledge with AI to enhance decision-making processes and improve outcomes in specialized domains.